

Guided Practice

Create a Simple Tic-Tac-Toe Game Using GitHub Copilot

Generative AI tools like **GitHub Copilot** act as an AI pair programmer—helping developers write, complete, and improve code faster. In this guided practice, you will see how Copilot can assist in building a small game with minimal effort.

Part 1: Creating a Simple Tic-Tac-Toe Game Using GitHub Copilot

Objective

Build a browser-based Tic-Tac-Toe game using:

- HTML
 - CSS
 - JavaScript
 - GitHub Copilot inside VS Code
-

Step 1: Project Setup

Create a folder:

```
tic-tac-toe/  
├── index.html  
├── style.css  
└── script.js
```

Open the folder in VS Code with GitHub Copilot enabled.

Step 2: Generate HTML with Copilot

In `index.html`, type this comment:

```
<!-- Create a Tic Tac Toe board with 9 cells and a reset button -->
```

GitHub Copilot suggests:

```
<!DOCTYPE html>
<html>
<head>
  <title>Tic Tac Toe</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

<h1>Tic Tac Toe</h1>

<div class="board">
  <div class="cell"></div>
  <div class="cell"></div>
  <div class="cell"></div>
  <div class="cell"></div>
  <div class="cell"></div>
  <div class="cell"></div>
  <div class="cell"></div>
  <div class="cell"></div>
  <div class="cell"></div>
</div>

<button id="reset">Reset</button>

<script src="script.js"></script>
</body>
</html>
```

Step 3: Generate CSS with Copilot

In `style.css`, type:

```
/* Style Tic Tac Toe board */
```

Copilot suggests:

```
body {
  text-align: center;
  font-family: Arial;
}

.board {
  display: grid;
```

```
    grid-template-columns: repeat(3, 100px);
    gap: 5px;
    justify-content: center;
}

.cell {
    width: 100px;
    height: 100px;
    background: #f2f2f2;
    font-size: 40px;
    display: flex;
    align-items: center;
    justify-content: center;
    cursor: pointer;
}
```

Step 4: Generate JavaScript Logic

In `script.js`, type:

```
// Add game logic for Tic Tac Toe
```

Copilot suggests:

```
const cells = document.querySelectorAll(".cell");
let currentPlayer = "X";

cells.forEach(cell => {
    cell.addEventListener("click", () => {
        if (cell.textContent === "") {
            cell.textContent = currentPlayer;
            currentPlayer = currentPlayer === "X" ? "O" : "X";
        }
    });
});

document.getElementById("reset").addEventListener("click", () => {
    cells.forEach(cell => cell.textContent = "");
    currentPlayer = "X";
});
```

Step 5: Add Winner Detection (Ask Copilot)

Add comment:

```
// Add function to check winner
```

Copilot suggests:

```
function checkWinner() {
  const winPatterns = [
    [0,1,2],[3,4,5],[6,7,8],
    [0,3,6],[1,4,7],[2,5,8],
    [0,4,8],[2,4,6]
  ];

  winPatterns.forEach(pattern => {
    const [a,b,c] = pattern;
    if (
      cells[a].textContent &&
      cells[a].textContent === cells[b].textContent &&
      cells[a].textContent === cells[c].textContent
    ) {
      alert(cells[a].textContent + " wins!");
    }
  });
}
```

Call `checkWinner()` after each move.

Outcome

You now have a **fully working Tic-Tac-Toe game** created largely with GitHub Copilot assistance.

Part 2: GitHub Copilot at Shopify (Industry Example)

Background

Shopify has thousands of developers working on large-scale commerce platforms. They adopted GitHub Copilot to:

- Speed up feature development
 - Reduce boilerplate coding
 - Improve developer satisfaction
-

How Shopify Uses GitHub Copilot

- Generate React components
 - Suggest GraphQL queries
 - Auto-complete tests
 - Refactor legacy code
-

Benefits Observed

- Faster coding
 - Reduced repetitive work
 - Developers focus on business logic
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Impact (Illustrative)

Metric	Before Copilot	After Copilot
Feature dev time	High	Reduced
Boilerplate writing	Manual	Auto
Developer satisfaction	Medium	High

Quick Check

Q1. What is GitHub Copilot?

- ✓ AI pair programmer

Q2. Which languages were used in Tic-Tac-Toe?

- ✓ HTML, CSS, JavaScript

Q3. True/False

Copilot replaces developers.

- ☐ False

Q4. Name one benefit Shopify gained.

- ✓ Faster development
-

Summary

GitHub Copilot enables developers to:

- Build apps faster
- Reduce manual effort
- Improve productivity

This guided practice shows how GenAI can be applied in **real coding scenarios**.